Adjusting AC frequency restores balance （ inertia makes the average AC effect balance it）. Find $\sqrt{\frac{k_e}{k_m}}$. 【Double lines are rigid insulating rods; single lines, wires; curves, flexible massless wires. Ignore wires' magnetic effects on M, N and electric and magnetic fringing】

VII. （20 points）

Two identical light metal springs of stiffness k hang vertically from a horizontal insulating bar, joined to a rigid uniform metal rod: mass m, length l. Length equals their spacing. Uniform field B is inward. Rod, springs and AC source u form a loop, Fig. （a）. Resistance and the loop field relative to B are negligible. In Fig. （b）, an inductor and capacitor are connected in parallel to the same u. If source currents always match, express C, L of （b） using （a）'s known device parameters.

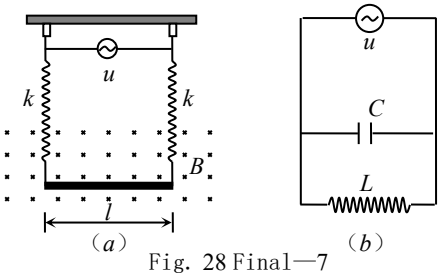


Fig. 28 Final—7

VIII. （20 points）

Steven Chu and two others won the 1997 Nobel in Physics for trapping neutral atoms. These questions explain this trapping. （Note: atomic parameters are averages over many atoms or repeated identical processes of one atom）.

（1） A ground-state atom at rest absorbs photons at v_0 resonantly, reaching its first excited state, Fig. （a）. Atoms move thermally; take speed v_0. A laser meets it head-on. Find its resonance frequency v and the speed change due to absorption. （$hv_0 \ll mc^2$, m is the atomic mass, $h=6.63\times10^{-34}$J•s）.

（2） Absorption is instantaneous, but the excited atom does not return immediately to ground state. Its stay in the excited state is that level's mean lifetime. The first excited state's mean lifetime is τ. On returning to ground state, it at once absorbs another head-on laser photon. Find the mean acceleration between the instants just before successive absorptions. Photons are equally likely in all directions. Very frequent collisions make its mean momentum change zero.

（3） The atom starts at v_0 along positive z; a laser meets it along negative z, causing resonance. At fixed laser frequency, repeated absorption must slow the atom to rest. It crosses a nonuniform field $B(z)$ for trapping, Fig. （c）. Interaction with the field changes the excited-state energy, as in Fig. （b）. At field B, an energy level E becomes $E+\Delta E$, where $\Delta E=\mu B$, μ is a known constant. Find B as a function of z.

（4） Lithium: $m=1.0\times10^{-26}$kg, initial $v_0=1.2\times10^{3}$m • s^{-1}; rest resonance frequency